

CS348 Computer Networks Quiz-2, Fri 12 Sep 2025, 08.35am-09.20am, Max. Marks: 10

Name: SAMPLE ANSWERS Roll number: _____ House: _____

General instructions

- Write only in the space provided. Answer briefly but crisply. Write neatly and clearly.
 - You are allowed to refer to your own hand-written notes only.
 - All answers have to be (briefly) explained. State any necessary assumptions.
1. [2+1+2=5 marks] A sysad Mnzncrfe Cdconisn has to manage 4 departments D0, D1, D2, D3, which need to support 500, 500, 100 and 100 hosts respectively. She got her ISP to give her some blocks of addresses, specifically: 194.1.0.*, 194.1.1.*, 194.1.2.*, 194.1.3.*, 194.1.4.*, 194.1.5.*
- (a) Help Cdconisn allocate the above IP prefixes to the 4 departments so as to minimize the number of entries in the routers within her institute? Specify the subnet(s) of each department. Allocate so that she can return one of the above 6 blocks, to save some money for the institute.

→ 512-2 addresses available
194.1.0.0/24 } → 194.1.0.0/23 → D0
194.1.1.0/24 }
→ 512-2 addresses available
194.1.2.0/24 } → 194.1.2.0/23 → D1
194.1.3.0/24 }
→ 194.1.4.0/25 (128-2 addresses) → D2
194.1.4.0/24 } → 194.1.4.128/25 (128-2 addresses) → D3
194.1.5.0/24 can be returned

- (b) Can all communication within department D0 happen without a layer3 router? Justify (< 30 words).

Yes, it is a single subnet

- (c) Cdconisn heard about CIDR and wants to know how the other routers in the Internet will represent her company in their routing table. Help her. (Max 3-4 lines).

D0 + D1 can be combined

194.1.0.0/23 } → 194.1.0.0/22
194.1.2.0/23 }

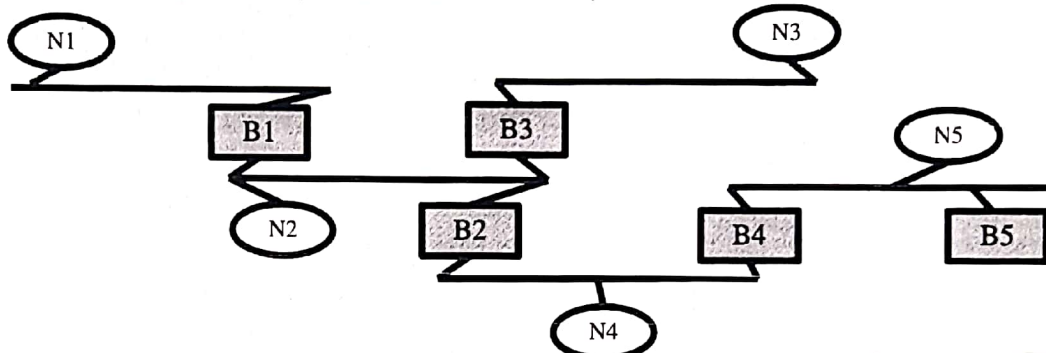
D2 + D3 can be combined back to 194.1.4.0/24

Far away routers will have above 2 entries

The 2 entries cannot be further combined

2. [5 marks] Prof Likbasr Canamgal deploys the bridged ethernet below with 5 learning bridges. However the professor installed a buggy code on B2 due to which it does not learn anything. They all start with empty forwarding tables. Draw the forwarding tables at each of the 4 non-buggy bridges, after each of the following packet transmissions reach their respective receivers. A total of 12 tables – draw in pencil, organize neatly! Explain briefly. You may denote the two interfaces of each bridge as *top-T* and *bottom-B* respectively (this notation is not marked in the diagram).

Transmission-1: N1 to N4, Transmission-2: N3 to N1, Transmission-3: N4 to N3



After T1
which is
flooded throughout

B1

N1	top
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After T2

B2 floods → B4
B4 does not flood

N1	top
N3	bot

After T3

B4 does not find

N4	bot
----	-----

B3

N1	bot
----	-----

N1	bot
N3	top

→ + =

N4	bot
----	-----

B4

N1	bot
----	-----

N1	bot
N3	bot

→ + =

N4	bot
----	-----

B5

N1	top
----	-----

N1	top
----	-----

N1	top
----	-----